

3D Nanolithography via Holographic Multi-Focus Metalens

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3D nanolithography based on two-photon polymerization (TPP) allows for the high-precision fabrication of nearly arbitrary 3D micro/nanostructures, finding extensive applications in areas such as micro-optics, micro-mechanics, and biomedicine. However, the large size, complexity of optical systems, and high costs have significantly constrained the widespread adoption of 3D nanolithography technology in both scientific research and industry. In this study, a metasurface is introduced, for the first time, into 3D nanolithography resulting in the construction of a miniaturized and simplified TPP system that achieved efficient multi-focus parallel processing with high uniformity. A microlens array is fabricated, showcasing the system's application capacity to generate an array of devices with high consistency and quality. It is believed that the utilization of metasurface devices will provide a novel TPP operating platform, enabling richer and more flexible printing functionalities while maintaining system miniaturization and low cost.

1. Introduction

Additive manufacturing represents a highly advanced processing method for micro/nano functional structures and devices.^[1–4] 3D nanolithography based on two-photon polymerization (TPP) stands as a representative technology in this field, widely recognized for its true 3D manufacturing capability and high printing resolution beyond the optical diffraction limit,^[5–8] which enables the high-precision fabrication of metamaterials,^[9,10] sensors,^[11,12] microrobotics,^[13–15] and nanowrinkled structures.^[16] In addition to the ultimate goals of high resolution and high throughput, miniaturization and low cost represent the recent trend in the advancement of TPP technology,^[17,18] which aims to reduce entry barriers and facilitate the widespread adoption of 3D nanolithography technology. This is especially critical for modern application fields such as in situ manipulation,^[19] lab-on-fiber,^[20] and aerospace, where there

is a demand for compact, lightweight, and cost-effective optical systems.

Nevertheless, current TPP technologies are associated with intricate and expensive optical systems.^[21–28] For example, a standard single-focus TPP system typically comprises an objective lens, a beam expansion system, and other necessary optical components, while a multi-focus TPP system for increasing processing throughput requires additional functional elements to be added to the optical system, such as a diffractive beam splitter or a microlens array (MLA). Although there have been reports of simplified nanoprinters,^[17,29,30] challenges including optical path accumulation, size, and weight, still have to be resolved to facilitate the further

integration of TPP systems. As a planar optical device, metasurface aligned with the development trend of pursuing compact and integrated optical systems,^[31] such as spectrometer,^[32] Hartmann–Shack wavefront sensor,^[33] two-photon microscopic imaging,^[34] chip-scale metalens microscope,^[35] orbital angular momentum demultiplexer, and decoder,^[36] laser direct writing,^[37] fluorescence imaging,^[38] Meanwhile, it shows the powerful modulation ability of various optical parameters (such as phase, amplitude, polarization, wavelength, etc.) by elaborately designing the parameters of meta-atoms, such as material, structure, arrangement, etc.^[39–47] However, research on integrating advanced optical element metasurfaces into TPP systems is presently limited. The application of metasurfaces into TPP may lead to be cost-effective and hold substantial development potential. It is expected to simplify optical setups while achieving richer and more flexible printing functionalities.

Here, we proposed an ultracompact metasurface-based TPP (M-TPP) technique compared with traditional optical setups. As an illustration, in this study, metasurface was employed to replace various light field modulator components in a multi-focus TPP system, including a beam splitter, a dispersion compensating system, a beam expanding system, and an objective lens, which realized seven-foci parallel processing with high uniformity. The designed metasurface was specifically a multi-focus metalens (MFM) with a high numerical aperture (NA). Several different arbitrary models from 2D to 3D were printed to demonstrate the practicability of M-TPP. In addition, an MLA device with consistent focusing and imaging quality was fabricated, showcasing the application prospects of M-TPP. The compact and cost-effective M-TPP technique is expected to significantly simplify the optical setup of micro/nano additive manufacturing systems, which

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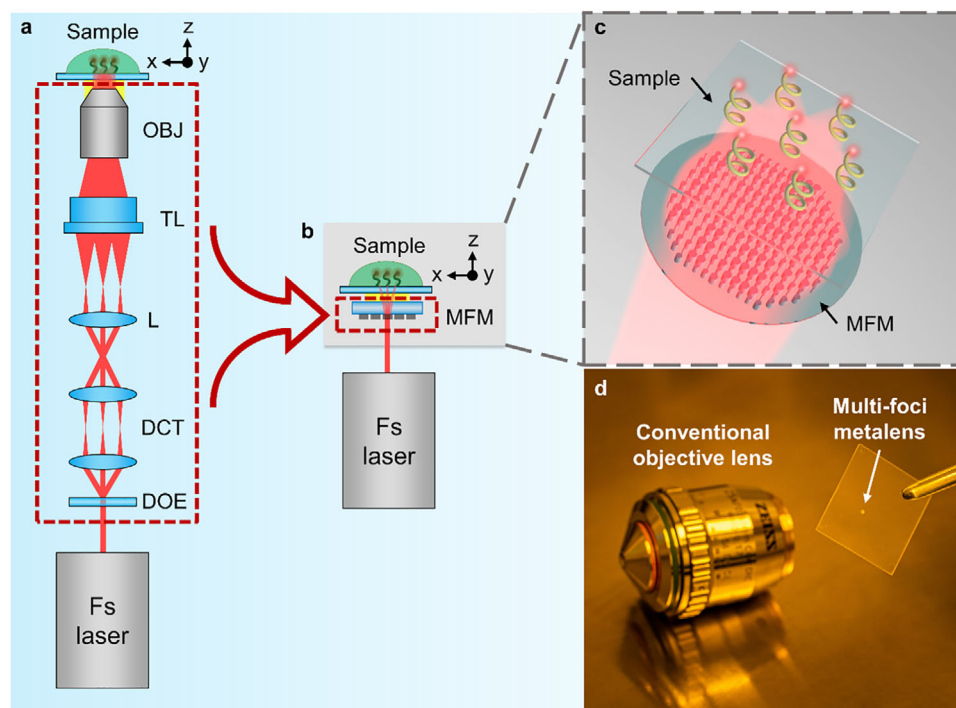


Figure 1. Design concept of M-TPP. a) Schematic diagram of traditional multi-focus TPP system based on a DOE. A DOE was used as a beam-splitting device to modulate incident light into multiple beams. Then femtosecond (fs) laser went through a DCT to correct spatial broadening of the diffracted beamlets and traveled to a 4f beam expanding system (lens (L) and tube lens (TL)) to make full use of the NA of the objective lens (OBJ). Finally, each beam was focused inside the photoresist by an objective lens, and complex 3D models were printed in parallel by moving a 3D stage. b) Schematic diagram of multi-focus TPP system based on the MFM. c) Printing process of 3D micro/nanostructure array via the MFM. d) Contrast diagram of the conventional objective lens and the MFM.

is expected to lead to widespread applications such as imaging, optical storage, advanced sensing, and optical encryption.

2. Results

The design concept of M-TPP is illustrated in **Figure 1**. Compared to commonly used light field modulator devices (such as diffractive optical element (DOE), digital micro-mirror device, spatial light modulator, etc), metasurfaces offer significant advantages such as the miniaturization platform and versatile integration, which forms the basis for constructing simplified and multi-functional TPP systems. Take the traditional multi-focus TPP system based on a DOE as an example (**Figure 1a**), a beam splitter DOE, a dispersion compensating telescope (DCT), a 4f beam expanding system, and an objective lens are necessary. In this study, a metasurface element MFM was utilized to replace all these optical components above, which greatly simplified the complexity of the existing system and was simple to operate, as shown in **Figure 1b**. The dimension occupied by the replaced optical components and their required optical path length was approximately several centimeters transversely and tens of centimeters axially. In contrast, the dimension of the MFM was only ≈ 2.2 cm $\times$ 2.2 cm transversely and ≈ 0.17 cm axially. Meanwhile, the total weight of the replaced optical components was on the order of kilograms, while the weight of the MFM was only 0.00025 kg. It is worth noting that the calculations for the dimension and weight of the MFM here include the glass substrate. The actual

dimension and weight of the meta-atoms responsible for modulation will be even lower. **Figure 1c** shows the printing process of an array of 3D micro/nanostructures using a MFM, increasing processing throughput by seven times compared to conventional single-focus manufacturing. And the morphological uniformity among structures was high. Furthermore, **Figure 1d** shows the comparison between the planar optical device metalens and the conventional objective lens. In comparison to the conventional objective lens, the MFM exhibits smaller size and weight, and the manufacturing cost is expected to be substantially lower, benefiting from advancements in deep-ultraviolet immersion lithography and wafer-scale nanoimprint lithography,^[48] as well as in inverse design method.^[49]

Based on the generalized laws of reflection and refraction,^[50] metalenses can rearrange the light wavefront and its propagation through the delicate design and arrangement of numerous subwavelength meta-atoms with special optical response characteristics.^[51,52] Furthermore, MFMs have been realized due to the capability of complete light field control,^[53–57] enabling focusing at different spatial positions along the longitudinal or transverse axes. Here, we employed the holography principle,^[58] to simultaneously encode the phase distributions of foci at different positions onto a single metalens element. In contrast to metalens arrays,^[53–55] this shared aperture design can achieve a longer work distance while maintaining the large NA and element size, which is beneficial for printing 3D structures with large heights. Additionally, this approach provided high design

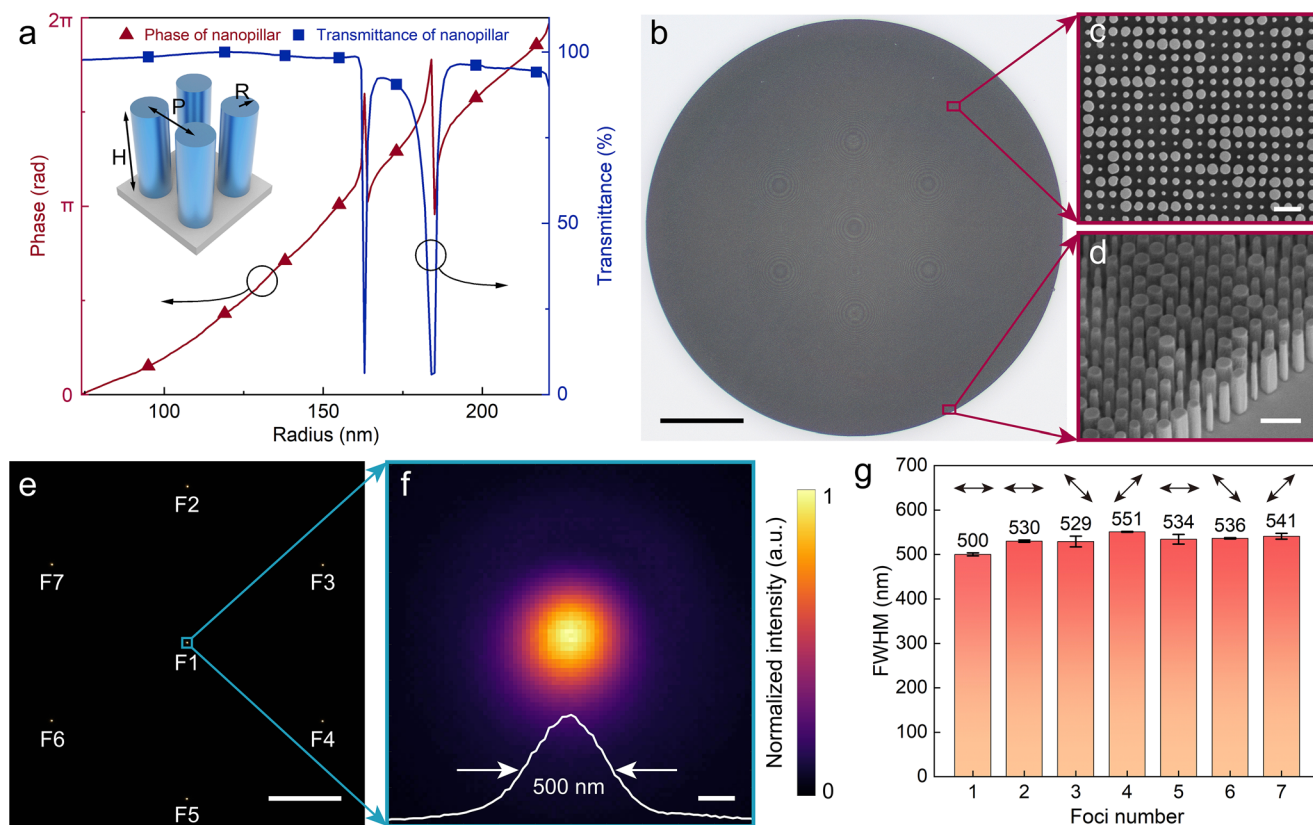


Figure 2. Design and focusing characterization of the MFM. a) The characterization of phase response and amplitude transmittance of SiN_x nanopillars as functions of nanopillar radius at a wavelength of 780 nm. The illustration is a geometrical diagram of SiN_x nanopillars. b) Optical microscope image of the MFM. Scale bar: 200 μm. c) Top view and d) oblique 45° SEM images of the MFM. Scale bar: 1 μm. e) Measured focusing results for seven foci of the MFM. Scale bar: 100 μm. f) Normalized intensity profiles and 1D normalized intensity distributions corresponding to F1. Scale bar: 0.2 μm. g) The measured FWHM of seven foci. The arrows indicated the calculation direction of 1D normalized intensity distribution (see Note S6 and Figure S8, Supporting Information).

flexibility, allowing for the arbitrary design of focus positions, amounts, uniformity, and arrangement when laser power is sufficiently high. The phase distribution of the MFM can be expressed as

$$\phi = \arg\left[\sum_{i=1}^N W_i \exp(j\phi_i)\right] \quad (1)$$

where W_i and ϕ_i are weight factor and phase distribution (see Note S1, Supporting Information) of the i^{th} focus of the MFM, the intensity ratio between foci was controlled by W_i to make them uniform, and then high printing consistency was obtained. According to the uniformity formula (see Note S2, Supporting Information), the uniformity of pre-design normalized total intensity was 99.6%. The refractive index of the propagation medium for the MFM was set to 1.518. And the focal length of the MFM was 594 μm. By subtracting glass heights, the working distances corresponding to the dip-in laser lithography (DiLL) working mode 1, DiLL working mode 2, and Oil working mode are ≈424, 339, and 339 μm (see Note S3 and Figure S2, Supporting Information), respectively. Due to the limited axial travel of the piezo stage, the maximum addressable print height in both modes is 300 μm. This can be increased by replacing the stage with a larger

axial travel, thereby fully utilizing the working distance of the MFM. Particularly, given the spectral bandwidth of fs laser (see Figure S3, Supporting Information), simulation results indicated that the absence of specific achromatic design for MFM does not significantly affect its focusing characteristics and efficiency, as illustrated in Note S4, Figure S4, and Table S1 (Supporting Information).

In the design of the MFM, phase retardations were achieved via polarization-independent SiN_x nanopillars based on transmission phase modulation, as shown in Figure 2a. The reason for the use of SiN_x was that it is almost transparent to laser at a wavelength of 780 nm and has a refractive index close to 2, much higher than normal glass materials. The finite-difference time-domain method was used to simulate and finally determine that the height (H) of the SiN_x nanopillars was 1150 nm, the period (P) was 520 nm, and the radius (R) was 95, 119, 138, 155, 173, 198, and 217 nm, respectively. Almost uniform seven-level phase modulation can be achieved under the premise of high amplitude transmission efficiency. Subsequently, the MFM was obtained by combining the established nanopillars library with the calculated phase distribution, which was fabricated through a combination of electron-beam lithography and inductively coupled plasma etching (see Note S5 and Figure S5, Supporting

Information). Figure 2b–d presents the fabricated results of the MFM with a diameter of ≈ 1.04 mm, where the equivalent NA for each focus was 0.8 (see Note S1, Supporting Information).

We define the transmission efficiency as the ratio of the optical power at the back of the MFM substrate to the incident optical power ($\frac{P_2}{P_1} \times 100\%$, see Figure S6 (Supporting Information), which was 80.13%. Meanwhile, the modulation efficiency was 73.28%, defined as the optical power at the back of the MFM substrate minus the zero-order optical power in the far field, divided by the incident optical power ($\frac{P_2 - P_3}{P_1} \times 100\%$, see Figure S6 (Supporting Information)). Because in metalenses, not all transmitted light can be focused. Some incompletely modulated light may be diffracted to undesired angles, which was mainly zero-order light in the center. The high efficiencies ensured that the power at its seven foci exceeded the polymerization threshold. We utilized a microscopic system to characterize the focusing performance of the MFM (see Figure S7a, Supporting Information). Figure 2e shows measured focusing results for seven foci, we name them focus one to focus seven (F1–F7) for convenience. According to the focusing direction of different foci, 1D normalized intensity distribution was reasonably obtained (see Note S6 and Figure S8, Supporting Information) to calculate full width at half maximum (FWHM). Figure 2f displays normalized intensity profiles and 1D normalized intensity distributions corresponding to F1. The measured FWHM of seven foci were 500, 530, 529, 551, 534, 536, and 541 nm, respectively, with a uniformity of 95.15%, as shown in Figure 2g. The experimental focusing results for seven foci of the MFM are shown in Figure S9 (Supporting Information). In addition, the uniformity of normalized total intensity (see Note S7 and Table S1, Supporting Information) of each focus was 95.21%. The experimental results prove that the MFM has excellent focusing uniformity, suggesting that the MFM can maintain high parallel processing consistency (see Note S8 and Figure S10 in Supporting Information).

To investigate the ultimate processing capacity of M-TPP, a suspended nanowire was fabricated as shown in Figure 3a,b). The lateral and axial sizes of the thinnest linewidth were 100.8 and 159.7 nm ($112.9 \times \sqrt{2} = 159.7$), respectively. It demonstrates that M-TPP can fabricate 3D micro/nanostructures with sub-diffraction-limited feature sizes. Subsequently, 2D and 3D micro/nanostructure arrays with exceptional uniformity were printed to showcase the arbitrary models manufacturing capability of M-TPP. As shown in Figure 3c, we printed the WNLO (Wuhan National Laboratory for Optoelectronics) logo array, which exhibited a favorable appearance. The SEM images demonstrate that M-TPP excels in 2D model forming. Besides, we printed an Arabic numerals array, and numerals 1–9 within the white dashed box in Figure 3d were printed using a single focus. This confirms the hexagonal topological splice capability of M-TPP. Movie S1 (Supporting Information) displayed its real-time printing process. Furthermore, this splicing method was not restricted to printing only nine models using a single focus, it can be extended to printing continuous structures and any quantity of models using a single focus (see Figure S11, Supporting Information). 2.5D models of the flower and the Great Wall of China were printed as shown in Figure 3f–i. The detailed grain structure of the petals in Figure 3g demonstrates the capability of M-TPP to print intricate models with fine details. 3D models of

the bridge with cavity and microcantilever with overhanging part were printed as shown in Figure 3j–m, which demonstrates the 3D nanolithography capability of M-TPP.

We printed an MLA device with a hexagonal topology arrangement to verify the practical application potential of M-TPP. MLAs were widely used in optical sensors,^[59] integral imaging,^[60] and other fields due to their high integration and consistency. For analytical convenience, we selected the printing of a single microlens using a single focus as an example to fabricate an MLA. Figure 4a shows SEM images of the fabricated MLA. IP-S was used as the material of the MLA, which has a high proximity effect, low shrinkage, and is almost transparent to visible light. The illustrations show that the fabricated microlens had smooth surfaces and stable 3D structures. The measured diameters of each microlens are 58.68, 58.29, 58.34, 58.28, 58.30, 58.37, and 58.31 μm (the uniformity is 99.66%), closely matching the design diameter of 56 μm . We increased the laser power to ensure sufficient polymerization, resulting in the diameter measurement values being slightly larger than the design value, but this hardly affects the optical performance of the MLA. Figure 4b presents the focusing performance results of the MLA (the measurement system is shown in Figure S7b (Supporting Information), where a uniform focus array can be clearly seen. Subsequently, Figure 4c,d) shows the imaging photographs of the letter “H” by the fabricated MLA, achieving clear imaging functions. Figure S18 (Supporting Information) presents lateral 1D normalized intensity distributions corresponding to lenses 1–7, respectively. FWHM was 1.554, 1.575, 1.629, 1.606, 1.587, 1.58, and 1.609 μm , respectively, with a uniformity of 97.64%. Figure S19 (Supporting Information) presents axial 1D normalized intensity distributions corresponding to lenses 1–7, respectively. Focal length was 78, 78, 82, 82, 80, 82, and 78 μm , respectively, with a uniformity of 97.5%. The design focal length of the plano-convex lens is calculated to be ≈ 69 μm using the formula $f_{\text{MLA}} = \frac{RC}{n_{\text{MLA}} - 1}$, where the radius of curvature (RC) is 35 μm and the refractive index of IP-S (TPP solid) is 1.507 @ 780 nm. There are two main reasons why the measured value is slightly greater than 69 μm . On one hand, there are no specific parameter requirements, only to verify the optical performance of the MLA. Therefore, we did not optimize the surface shape in the fabrication process, resulting in a larger radius of curvature. On the other hand, measurement errors also contributed to this discrepancy. In short, the focusing and imaging results demonstrate a consistent optical quality of the fabricated MLA due to the excellent performance of M-TPP.

The above results prove that replacing a high-end objective lens with a metalens brings integration and functionality enhancement, but it also leads to some performance degradation. First, an objective lens can compensate for spherical aberration by rotating a single correction collar, while the MFM lacks flexibility for adjustment according to actual printing conditions. Second, an objective lens is typically designed with broadband achromatic capability, making it compatible with nonlinear optical characterization techniques such as coherent anti-stokes Raman scattering, enabling in situ monitoring of the printing process. However, the MFM only has a 10 nm bandwidth centered at 780 nm. It is challenging to integrate it with advanced nonlinear optical characterization techniques. Third, an objective lens

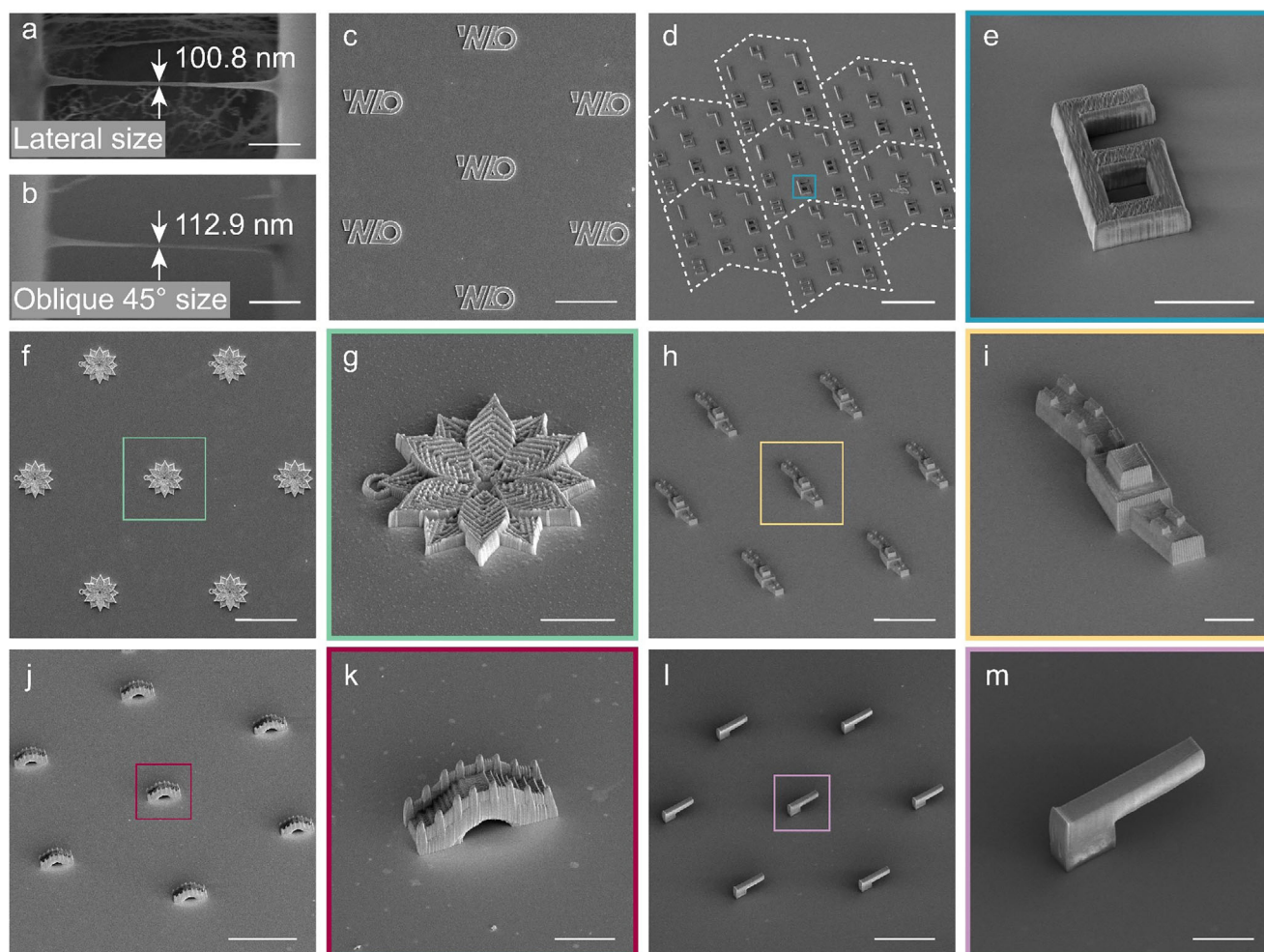


Figure 3. SEM micrographs of as-printed micro/nanostructures. a) Lateral and b) Oblique 45° views of the thinnest linewidth. c) SEM image of the WNLO logo array. d) SEM image of an Arabic numerals array. e) Zoomed-in SEM image of an Arabic numeral six corresponding to F1 printing. f) SEM image of a flower array. g) Zoomed-in SEM image of a flower corresponding to F1 printing. h) SEM image of the Great Wall of China array. i) Zoomed-in SEM image of the Great Wall of China corresponding to F1 printing. j) SEM image of a bridge array. k) Zoomed-in SEM image of a bridge corresponding to F1 printing. l) SEM image of a microcantilever array. m) Zoomed-in SEM image of a microcantilever corresponding to F1 printing. Scale bar in (a), (b): 1 μm ; scale bar in (c), (d), (f), (h), (j), (l): 100 μm ; scale bar in (e), (g), (i), (k), (m): 20 μm .

can achieve diffraction-limited focusing under light incident at different angles, thereby increasing printing throughput by using galvanometers. However, the MFM is currently designed only for normal incidence conditions, which makes it limited to the scanning mode based on a piezo stage. In the future, the performance of MFM can be further improved through tunable metasurface methods, advanced metalens chromatic aberration correction techniques, and optimization of the electromagnetic response characteristics of meta-atoms at different incident angles. This will advance the widespread application of metasurface platforms in TPP technology.

3. Conclusion

In conclusion, we have proposed and demonstrated a highly integrated and functional M-TPP technique. The metasurface provided a promising platform for the development of laser precision engineering which is not limited to the proposed MFM in

this work. It is expected to realize almost any desired light field modulation based on this platform, and the processing mode can be expanded from the scanning method to the plane projection method, and even the holographic projection method. Meanwhile, M-TPP can flexibly switch functions according to demands. If the functions of traditional systems need to be switched between different printing modes (such as multi-focus printing, projection printing, photoinhibition printing, etc.), it is often necessary to make numerous changes and recalibrations to the optical setup, which is both troublesome and time-consuming. In contrast, on the metasurface platform, it is easy to choose suitable printing functions by simply replacing metasurface elements with different functions, akin to the way commercial microscopes interchange objective lenses. It is believed that a highly integrated and versatile nanoprinter will be established based on the flat optical devices of metasurfaces, which can promote the widespread adoption of laser micro/nano 3D printing technology in scientific research and industry.

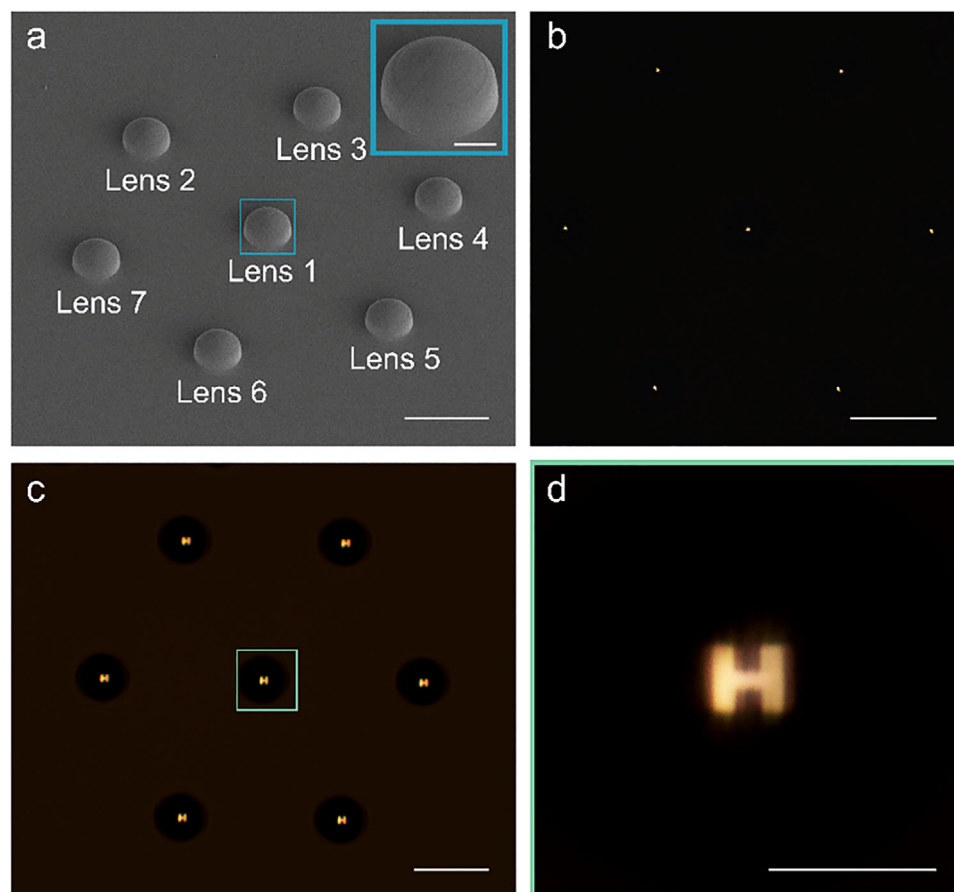


Figure 4. Optical characterization of the fabricated MLA. a) SEM image of the MLA. Scale bar: 100 μm . The illustration is a zoomed-in SEM image of a microlens corresponding to F1 printing. Scale bar: 20 μm . b) Focusing results of the MLA. Scale bar: 100 μm . c) Imaging results of the MLA. Scale bar: 100 μm . d) Zoomed-in imaging photograph of lens 1. Scale bar: 20 μm .

4. Experimental Section

Simulation of SiN_x Nanopillars: The optical properties of meta-atoms were simulated via the finite-difference time-domain method in the commercial software (Lumerical Inc.). The nanopillars were simulated using silicon nitride material on the glass substrate. The periodic boundary conditions were applied along the transverse direction. For the longitudinal direction, perfectly matched layer boundary conditions were utilized. A plane wave with the design wavelength of 780 nm was employed as incidence. Considering manufacturing constraints, the diameter and height of nanopillars were swept within a region to maximize the amplitude transmission efficiency while covering the phase modulation range of 2π .

Optical Setup of M-TPP: The optical setup of M-TPP is shown in Figure S20 (Supporting Information), the laser source of the system was a fs mode-locked Ti: sapphire laser (Coherent, Chameleon Discovery; 100 fs of pulse width, 80 MHz of repetition rate) with an output wavelength of 780 nm, an acoustic-optic modulator (AA OPTO-ELECTRONIC, MT110) was used to modulate laser power and on-off, a quarter-wave plate (Thorlabs, AQWP10M-580) was used to modulate laser to circular polarization state, a dichroic mirror (Thorlabs, DMLP650R) was used to combine excitation light and illuminate beam (Thorlabs, M617L5), a piezo stage (Physik Instrument, P-563.3CD) was used to move sample in XYZ direction. For suspended nanowires, a stage (Daheng Optics, GCM-T13M2L) driven by a motorized actuator (Thorlabs, Z812B) was used to print nanowires. The processing information was collected through an air objective lens (Olympus, UPLFLN; 0.30 NA, $\times 10$), a filter (Thorlabs, FESH0750) was used to

filter out excitation light, and a tube lens (Olympus, SWTLU-C) focused light onto CMOS camera (Daheng Imaging, MER-U3).

Performance Characterization of the MFM and the MLA: Optical setups that characterized the MFM and the MLA are shown in Figure S7 (Supporting Information), laser source, quarter-wave plate, tube lens, and CMOS camera were the same as above. For the MFM, the refractive index liquid (Cargille, Series A) was used as the propagation medium, and focus information was collected by an air objective lens (Olympus, UPLFLN; 0.30 NA, $\times 10$) and an oil-immersion objective lens (Motic, EC-H Plan, 1.25 NA, $\times 100$), respectively. For the MLA, an air objective lens (Olympus, UPLFLN; 0.30 NA, $\times 10$) was used for focus detection. The imaging experiment was performed by placing the object behind a microscope (Motic, BA410) illumination source (Motic, halogen bulb).

Printing Process of TPP: A glass substrate (Thorlabs, CG15CH2, and CG00C2) was ultrasonically cleaned with acetone (KESHI) and anhydrous ethanol (KESHI) for 10 min, respectively. The substrate was then soaked for 12 h in a solution composed of 2 mL of 3-(Trimethoxysilyl) propyl methacrylate (Bidepharm), 100 mL of anhydrous ethanol, and 5 mL of deionized water, and finally dried with a nitrogen gun as the printing substrate. Then a suitable working mode was chosen (see Note S3 and Figure S2, Supporting Information) based on the photoresist type for the printing process. After TPP processing, the sample was placed into propylene glycol methyl ether acetate (PGMEA; Aladdin) to develop for 20 min, then immersed in isopropyl alcohol (KESHI) for 3 min to wash PGMEA on the structure surface, and finally air dried naturally. For suspended nanowires, after development, the sample was immersed in anhydrous

ethanol (KESHI) and dried by a supercritical dryer (Tousimis, Samdri-PVT-3D) to avoid the collapse of suspended nanowires resulting from surface tension. 3D printing parameters for all structures are shown in Table S2 (Supporting Information).

Morphological Characterization of Micro/Nanostructures and the MFM: The morphologies of micro/nanostructures and the MFM were characterized using a field-emission SEM (FEI, Nova NanoSEM 450). The optical microscope image of the MFM was measured by a 3D laser scanning confocal microscope (Keyence, VK-X1100). The contrast diagram of the conventional objective lens and the MFM was taken with a full-frame mirrorless camera (Sony, A7 III) with a zoom lens (Tamron 28–200 mm F/2.8-5.6 Di III RXD).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

X.W., X.F., Y.L., K.X., F.C., X.Y., H.G., and W.X. are inventors on a patent application related to this work filed by Huazhong University of Science and Technology (no. CN116430678A, filed 23 March 2023). The authors declare that they have no other competing interests.

Data Availability Statement

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Information.

Keywords

3D nanolithography, compact optical system, metasurface, multi-focus metalens, two-photon polymerization

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